

LESSON 1

Descriptive Statistics

1. Basic concepts

2. Analysis of one-dimensional random variables

- 2.1. Frequency tables
- 2.2. Parameters of position
- 2.3. Parameters of dispersion
- 2.4. Parameters of shape
- 2.5. One-dimensional graphs

3. Analysis of two-dimensional random variables

- 2.1. Cross-tabulation table
- 2.2. Scatter plots
- 2.3. Covariance matrix
- 2.4. Linear correlation coefficient
- 2.5. Multiple Box-Whisker
- 2.6. Interpretation of relationships

Statistics

«The science whose aim is the collection and analysis of data through the use of mathematical models and software tools.»

«Science that studies the random phenomena»

Random

«Are random phenomena those that follow the laws of chance: unpredictability of the results and consistency in the same»

Descriptive Statistics

- Descriptive statistics allows us to organize and summarize the information contained in a set of observations (data).
- The goal is to extract the underlying information in the data

Inferential Statistics

- Tries to explain the random behavior of the experiment from the data.
- Contrasting hypothesis
- Make predictions

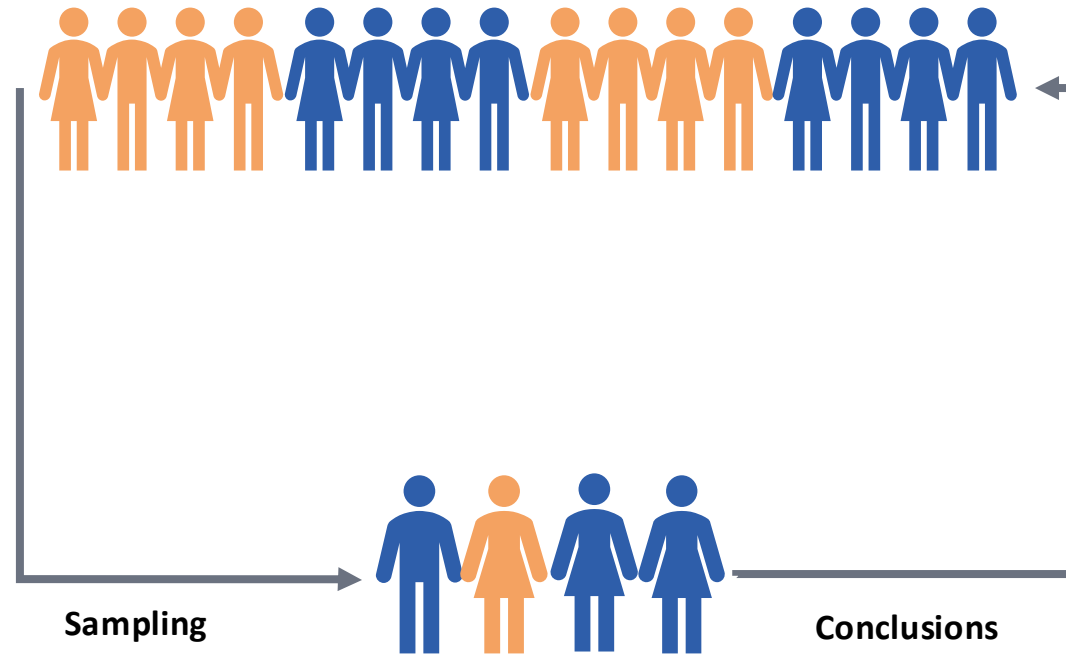
Basic concepts

Population

Set of objects that we are interested in, for which we intend to get conclusions.

Sample

Subset formed by parts of the objects (individuals) of a population.



The sample must be “representative” of the population. Only guarantee of “representativity”: **Random sampling.**

Random variable

- All real populations have variability. That is, it is not possible to have two identical pieces.
- A random variable is any characteristic, that can be expressed numerically, that fluctuates among the individuals of the population.

Data

- Data is the result or finding of an observation, measurement or analysis. There is no value or utility itself.

Information

- Information is already interpreted data. Has value and is useful.

Basic concepts

Data types

Quantitative data

The possible outcomes of the observations in the study are numeric.

Discrete

they take only certain values of the real line.

Continuous

they can take any value of the real line or a part of it.

Qualitative data

The possible outcomes are qualities. They describe some characteristic or quality observed in the study

Frequency tables

One-dimensional tables summarize the information contained in a dataset in a clear tabular format.

A frequency table shows the number of times each value, or each range of values, of a variable appears in the dataset. This count is known as the frequency.

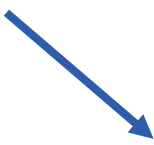
Frequency tables can be applied to:

- Qualitative data
- Discrete quantitative data
- Continuous quantitative data (grouped into intervals)

Frequency tables

Frequency tables for qualitative data

Injuries	Absolute frequency (Fi)	Relative frequency (fi)
Fall	12	0.154
Crushing	2	0.026
Blow	20	0.256
Cut	3	0.038
Abrasion	11	0.141
Burn	26	0.333
Others	4	0.051


$$f_i = \frac{F_i}{N}$$

Frequency tables

Frequency tables for discrete quantitative data

Nº of members in the household	Absolute frequency (F_i)	Cumulative frequency	Relative frequency (f_i)	Cumulative relative frequency
1	3	3	0.06	0.06
2	7	10	0.14	0.20
3	15	25	0.30	0.50
4	13	38	0.26	0.76
5	8	46	0.16	0.92
6	3	49	0.06	0.98
7	1	50	0.02	1.00
8	0	50	0.00	1.00

Frequency tables for continuous data

- If the variable is continuous existence of the variable field is divided into a set of K intervals, usually of the same length.
- It is not possible to determine a priori the optimal length of the intervals and, therefore the number of these which we will have to consider.
- Too many would lead to a table very verbose and difficult to interpret and a too small number would make us lose a very important part of the information contained in the data.
- Between **5** and **15 intervals**, depending on the number of data, would be an appropriate number.
- In general, and of guidance form, is used a number of intervals equal to the square root of the number of available data.

Frequency tables

Frequency tables for continuous data

Inner diameter	Absolute frequency (F_i)	Cumulative frequency	Relative frequency (f_i)	Cumulative relative frequency
1	0	0	0	0
2	0	0	0	0
3	3	3	0.03	0.03
4	22	25	0.22	0.25
5	49	74	0.49	0.74
6	19	93	0.19	0.93
7	6	99	0.06	0.99
8	1	100	0.01	1.00

Parameters of position – Average/Mean

It is the most used position parameter and corresponds to the arithmetic mean ($\bar{x}$) of the observed data:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \sum_{i=1}^n \frac{x_i}{n}$$

When data are grouped by frequency using the expresión:

$$\bar{x} = \frac{\sum_{i=1}^n x_i \cdot n_i}{\sum_{i=1}^n n_i}$$

Parameters of position – Average/Mean

Properties:

- All the values of the distribution are involved in their calculation by which each affects the average.
- The arithmetic mean is expressed in the same units as the variable.
- It is the center of gravity of the entire distribution, representing all of the observed values.
- It is unique.
- The algebraic sum of the deviations of individual values on the average is zero.
- In general, the average represents very well the "Center of gravity" of the distribution of data giving a clear idea about the location of the same. However, it is a **weak parameter given the existence of abnormally values** distinct to those who set up the "thickness" of the group. Thus, **if the distribution is asymmetric** must turn to other position parameter: the median.

Parameters of position – Median

The **median** ($\tilde{x}$) is the value that represents the middle of a dataset when the values are arranged in numerical order.

If the number of values is **odd**

$$\tilde{x} = \text{value in place } \frac{n+1}{2}$$

The median is the **single middle value**.

If the number of values is **even**

$$\tilde{x} = \text{mean value for } \frac{n}{2} \text{ and } \frac{n}{2} + 1$$

The median is **the average of the two middle values**.

Parameters of position – Median

Properties:

- The median is influenced by the core values and it is insensitive to outliers.
- In case of a asymmetrical distribution, the median is not displaced from the point of central trend.

Parameters of position – Mode

It is that value which corresponds to a relative maximum in its absolute or relative frequency when the data grouped them by frequency. In a set of data may occur that there is more than one value that meets this characteristic which may not be unique or even in some cases may not exist. It is not a parameter that has special interest.

Properties:

- Represents more elements than any other.
- No value is affected by the extreme values, but its calculation is doubtful for continuous data.
- The fashion for a frequency distribution of pooled data can not be calculated exactly, the value of Mode can be affected by the method of grouping of class intervals.
- Sometimes random intervenes significantly and makes a non-representative value is repeated frequently.

Parameters of position – Percentiles and Quartiles

- It is called P_i percentile value which, once ordered smaller to greater, has under it an $i\%$ of data.
- The P_{25} , P_{50} and P_{75} percentiles, also known respectively as the first quartile Q_1 , second quartile Q_2 and third quartile Q_3 , are particularly useful for establishing the position of the data. Quartiles enable a detailed analysis of the distribution and are normally used to locate a piece of data within the set.
- The second quartile or fiftieth percentile (P_{50}) corresponds to the median

$$P_{50} = Q_2 = \tilde{x}$$

- 50% of the observed values lie between the first and third quartiles.

Parameters of dispersion– Variance (S^2)

The most commonly used measure is the corrected variance, calculated as the average of the squared deviations of the data from their mean. It is calculated using the following expression:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

When data are grouped by frequency using the expresión:

$$s^2 = \frac{\sum (x_i - \bar{x})^2 \cdot n_i}{\sum (n_i - 1)}$$

Parameters of dispersion – Standard deviation (S)

To express the variability in the same units that the data or the average, commonly used **standard deviation** which is nothing more than the positive square root of the variance.

$$S = +\sqrt{S^2}$$

Properties:

- It is influenced by the value of each observation.
- As a result of considering quadratic deviation that places greater emphasis on extreme deviations in other deviations.

Parameters of dispersion – Coefficient of Variation (CV)

In some cases, variability is expressed using a **dimensionless** measure called the coefficient of variation (CV).

$$CV = \frac{s}{x} \cdot 100$$

Parameters of dispersion – Range

The **statistical range** shows the spread of a dataset. It is calculated as the difference between the **maximum value** and **the minimum value** in the data:

$$\textit{Range} = \max(X) - \min(X)$$

It indicates how widely the data are distributed, but it only considers the two extreme values and ignores the rest of the dataset, so it is **sensitive to outliers**.

Parameters of dispersion – Interquartile Range

When the **mean** is not an appropriate measure of central position, the **variance** is also not suitable as a dispersion parameter because it is based on deviations from the mean.

In these cases, the **interquartile range (IQR)** is more appropriate. It is defined as the difference between the **third quartile (Q3)** and the **first quartile (Q1)**.

$$IQR = Q_3 - Q_1$$

The IQR is a **robust measure of dispersion**, useful when the dataset contains outliers or anomalous values.

Parameters of shape

Parameters that compare real data distributions with the characteristics of a normal distribution (reference model) are called shape parameters.

Shape parameters describe the form of a distribution independently of its central tendency or dispersion. The two main shape parameters are the Skewness and the Kurtosis.

Parameters of shape – Skewness (CA)

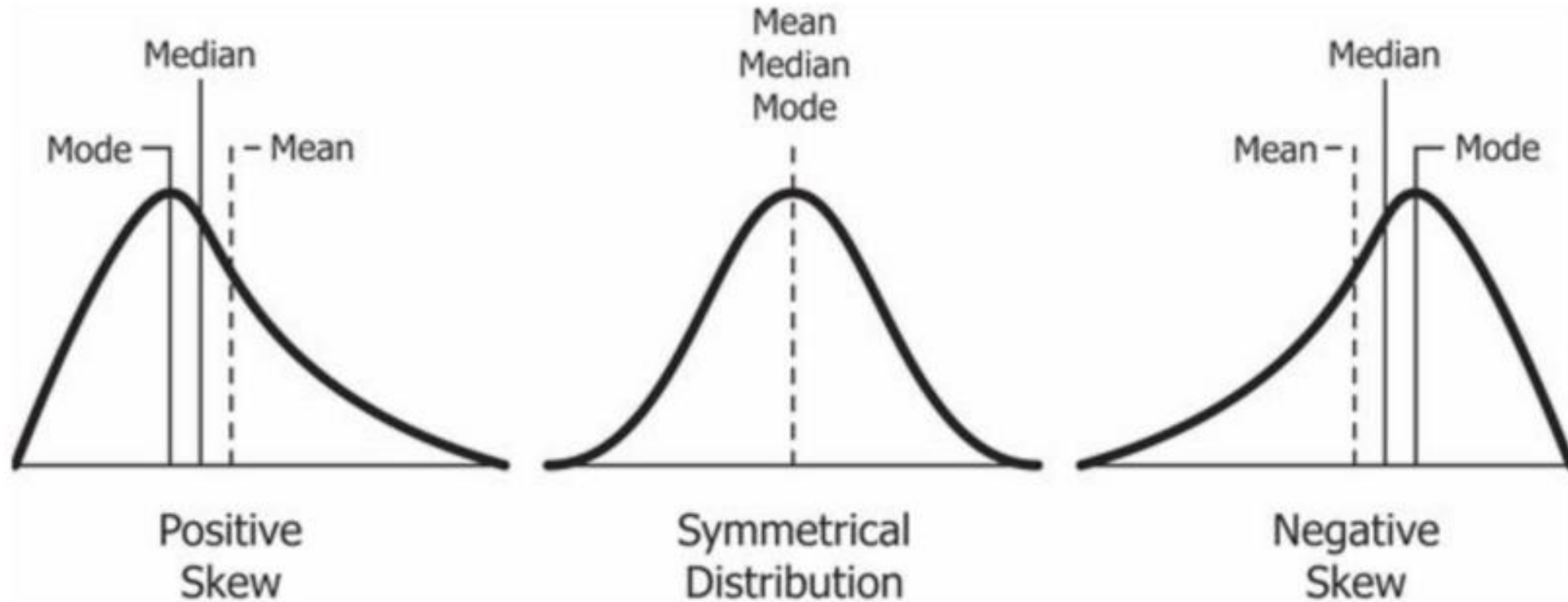
The skewness measures the **asymmetry** of the distribution.

$$CA = \frac{\frac{\sum_{i=1}^n (x_i - \bar{x})^3}{n - 1}}{S^3}$$

- If $CA < 0$ (negative skewness): longer tail to the left.
- If $CA = 0$: symmetric distribution (like the normal distribution).
- If $CA > 0$ (positive skewness): longer tail to the right.

If CA is in the range **[-2,2]** we can assume that **the variable follows a normal distribution.**

Parameters of shape – Skewness (CA)



Parameters of shape – Kurtosis (CC)

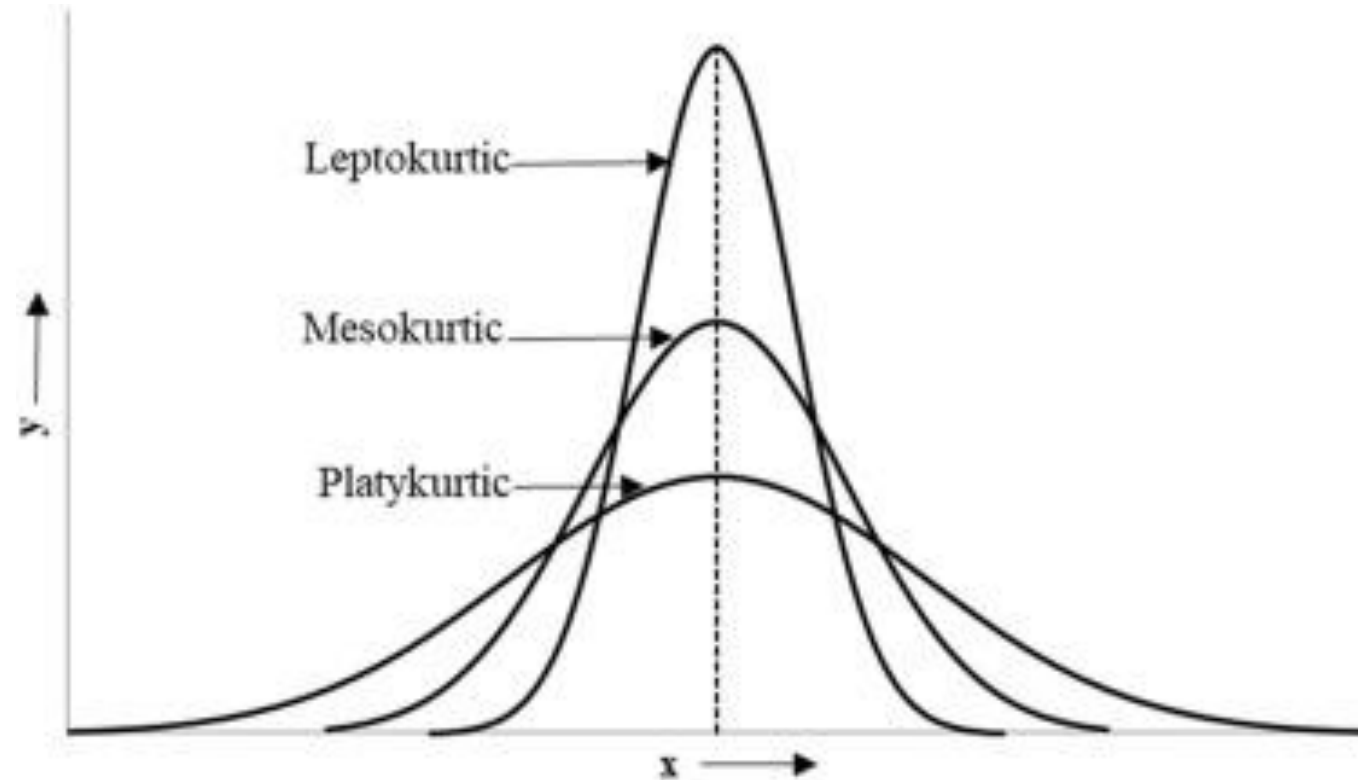
Measures the **degree of concentration of values around the mean** and the **weight of the tails** compared with a normal distribution.

$$CC = \frac{\frac{\sum_{i=1}^n (x_i - \bar{x})^4}{n - 1}}{S^4} - 3$$

- If $CC < 0$: platykurtic
- If $CC = 0$: mesokurtic
- If $CC > 0$: leptokurtic

If CC is in the range **[-3,3]** we can assume that **the variable follows a normal distribution.**

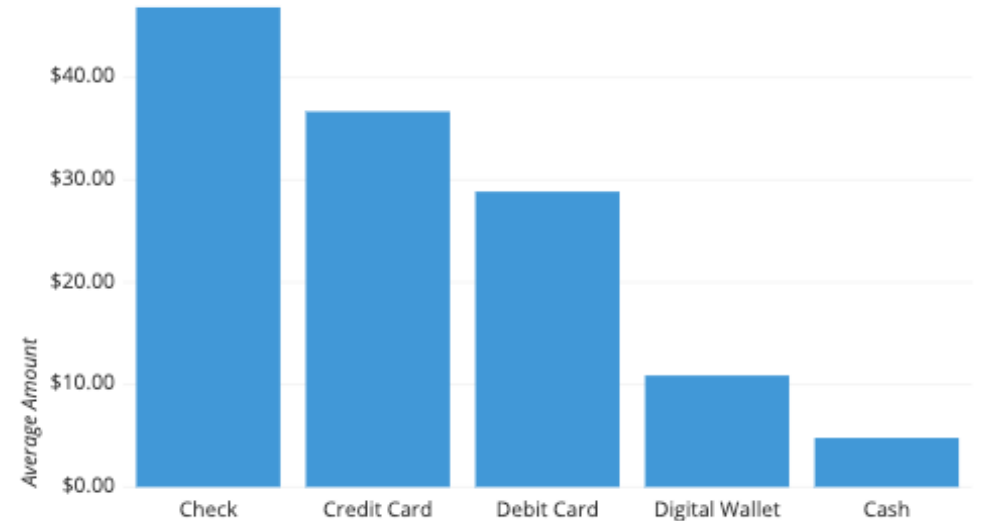
Parameters of shape – Kurtosis (CC)



One-dimensional graphs - Barchart

In descriptive analysis, data of qualitative or countable quantitative character are usually represented with a **bar chart**.

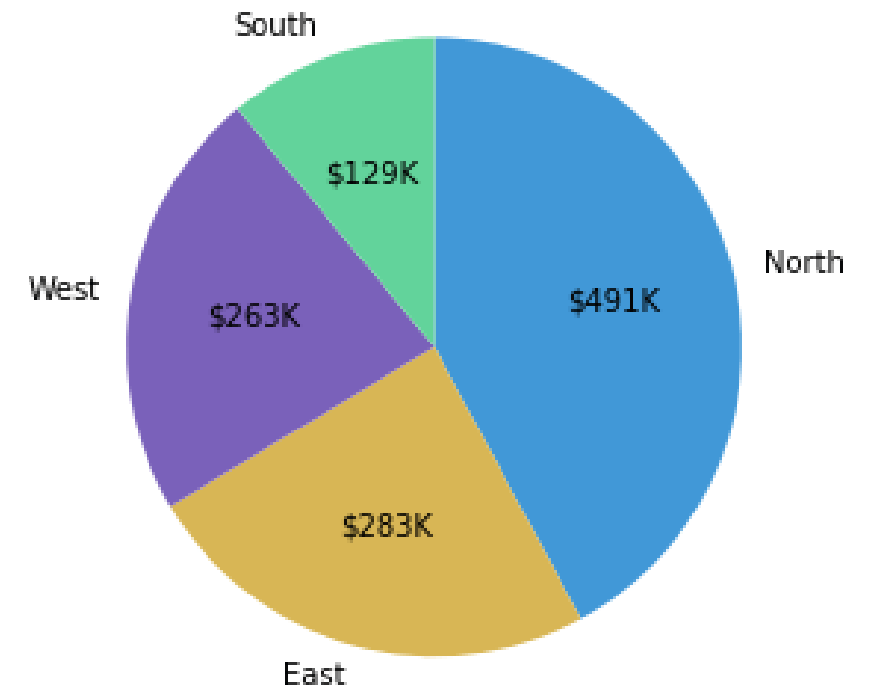
- In a bar chart, the different values or categories of a discrete qualitative or quantitative variable are shown on the horizontal axis.
- For each value of the variable, a bar is drawn with a height proportional to its **absolute or relative frequency**.
- It is a graphical way to represent the values obtained in preliminary frequency tables.



One-dimensional graphs - Piechart

Alternatively it is possible to build a piechart dividing the total area of a circle, in sectors of area proportional to the frequency of each of the alternatives in the case study.

- Usually used when there are few ways that represent (7 max).

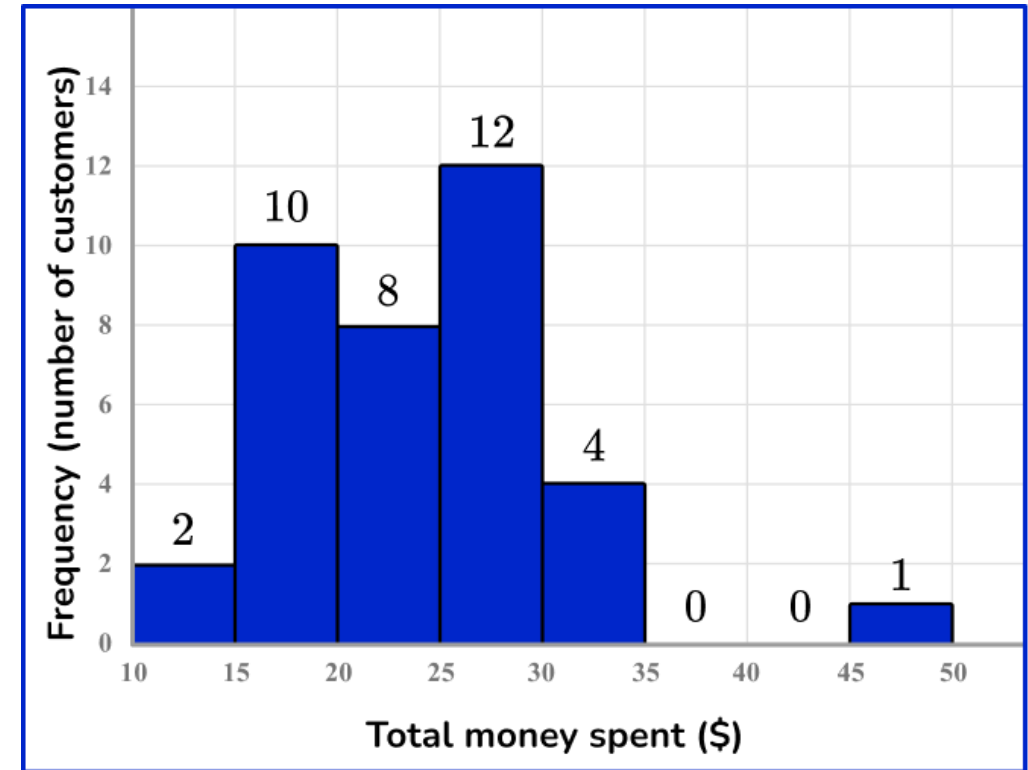


One-dimensional graphs - Histogram

Represent grouped frequencies of a continuous variable on intervals.

In the axis of ordinates (intervals) classes or the values of the variables created in such a way that the base of each rectangle represents the extent of the range.

Over each value rises a rectangle's height proportional to the absolute or relative frequency with which it is produced.



One-dimensional graphs - Histogram

Building a histogram

- 1) Not build histograms with less than 50 points.
- 2) Determine, approximately, the number of intervals as the square root of the number of data.
- 3) Calculate the amplitude approximate intervals as:

$$e = \frac{\max(X) - \min(X)}{k}$$

$$k = \sqrt{N} \text{ for } N \in [5,15]$$

- 4) Round up the extent of the range a suitable number.
- 5) Quantify the data falling in each interval

Building a histogram

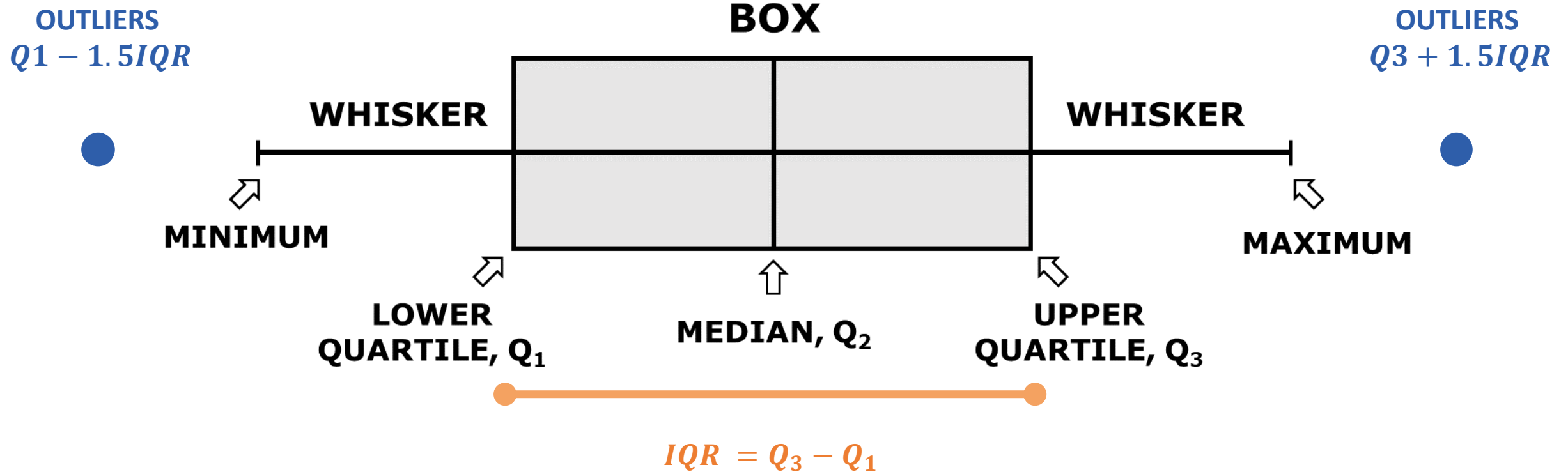
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One-dimensional graphs – Box-Whiskers



Analysis of two-dimensional random variables

Two-dimensional descriptive statistics **analyze pairs of variables** to understand how they are related. Each observation consists of two values, usually written as (X, Y) .

The goal is to describe the association between the variables, such as whether they increase or decrease together. Common tools include scatter plots, covariance, and correlation, which measure the direction and strength of the relationship between the two variables.

Both characteristics can be:

- Both quantitative
- Both qualitative
- One quantitative and one qualitative

Cross-tabulation table (both qualitative)

Estatura Peso	145 155	155 165	165 175	175 185	185 195	Total por Fila
40 55	9 75.0%	17 44.7%	0 0.0%	0 0.0%	0 0.0%	26 20,00%
55 70	3 25.0%	18 47.4%	31 53.4%	5 29.4%	0 0.0%	57 43,85%
70 85	0 0.0%	3 7.9%	24 41.4%	12 70.6%	3 60.0%	42 32,31%
85 99	0 0.0%	0 0.0%	3 5.2%	0 0.0%	2 40.0%	5 3,85%
Total por Columna	12 9,23%	38 29,23%	58 44,62%	17 13,08%	5 3,85%	130 100,00%

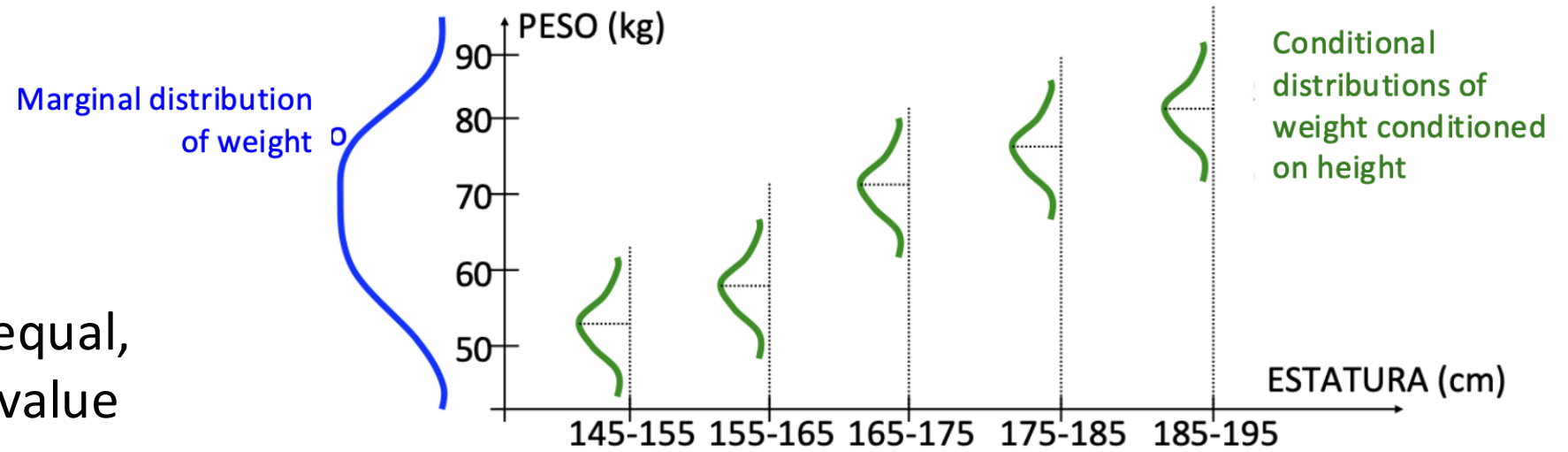
Marginal frequency of weight

Marginal frequency of height

Relative frequency of weight conditioned on height

Cross-tabulation table (both qualitative)

If all conditional distributions are equal, regardless of the value taken by the conditioning variable, the two random variables are **INDEPENDENT**.

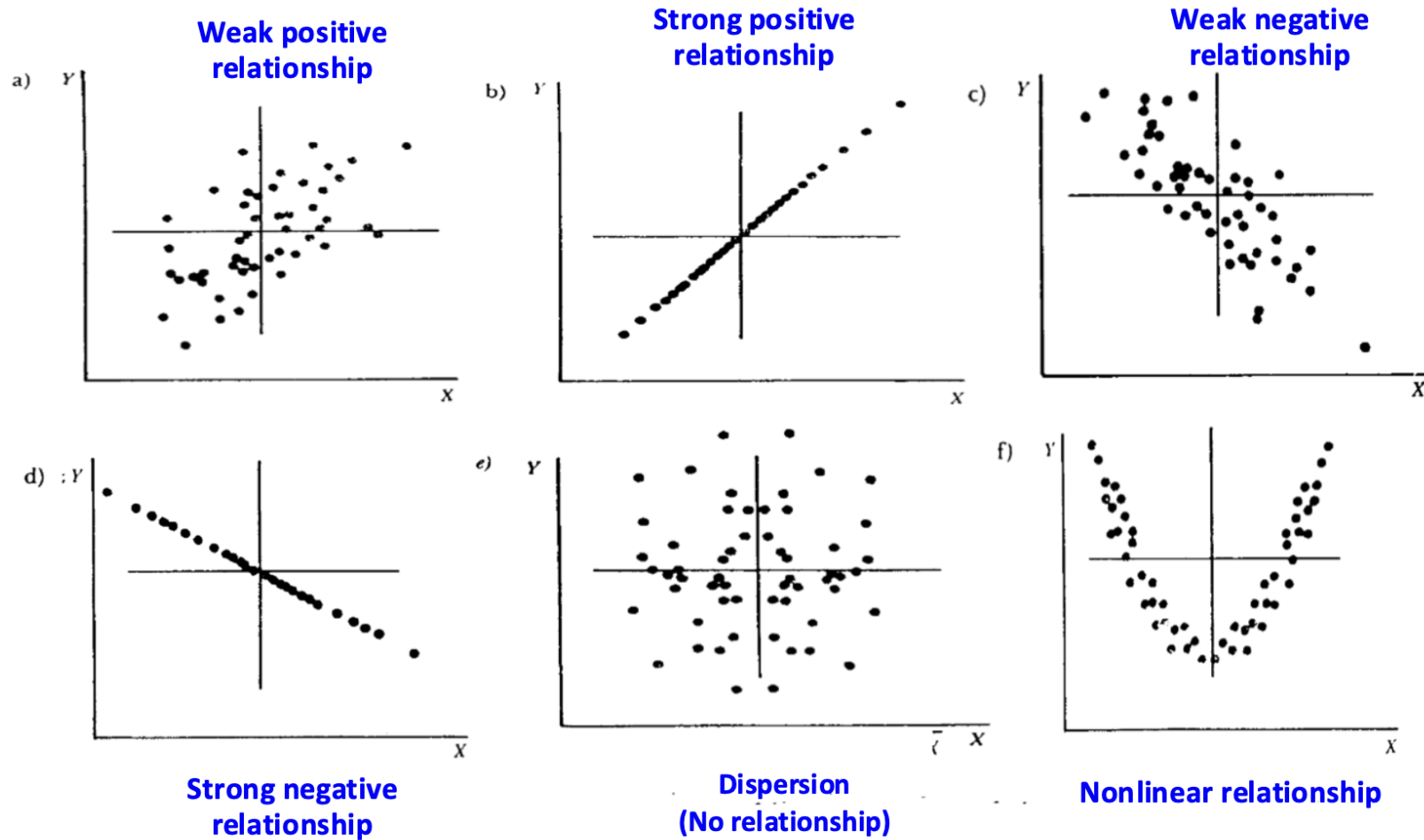


Marginal frequency of weight

Marginal frequency of height

Relative frequency of weight conditioned on height

Scatterplots



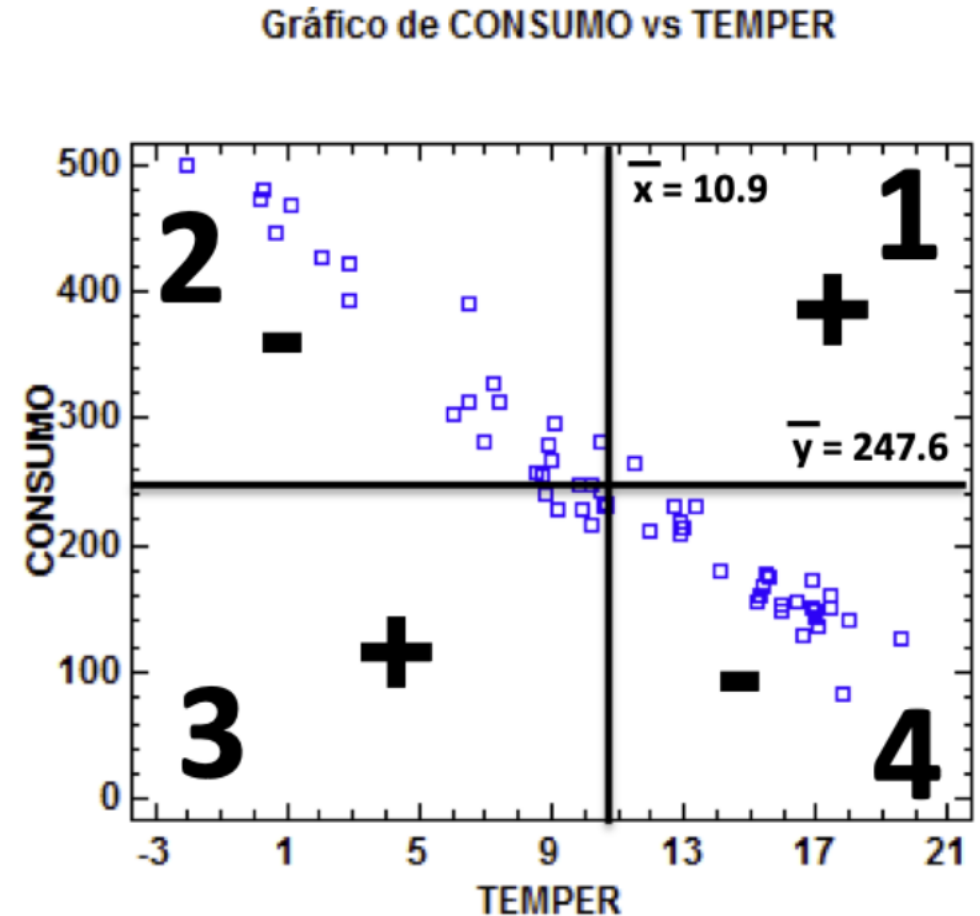
Covariance matrix

The scatter diagram of the relationship between CONSUMPTION and TEMPER indicates the sign (+ or -) of the product

$$(x_i - \bar{x})(y_i - \bar{y})$$

in each quadrant defined in the plane by the straight **lines $x = \bar{x}$ and $y = \bar{y}$** .

It can be seen that in most cases the sign is -, so the mean value of these products will be negative.



Covariance matrix

The units are the product of those of X by those of Y.

$$COV_{X,Y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

The value of the covariance does not allow us to form an idea of the degree of linear relationship that exists (since **it depends on the units of measurement**).

The covariance matrix:

$$\bar{V} = \begin{pmatrix} \sigma_1^2 & cov_{12} \\ cov_{21} & \sigma_2^2 \end{pmatrix}$$

Linear correlation coefficient

To overcome the covariance units problem, the linear correlation coefficient is used as a measure of the degree of linear relationship between two variables.

The linear correlation coefficient is calculated by dividing the covariance by the product of the standard deviations of the two variables:

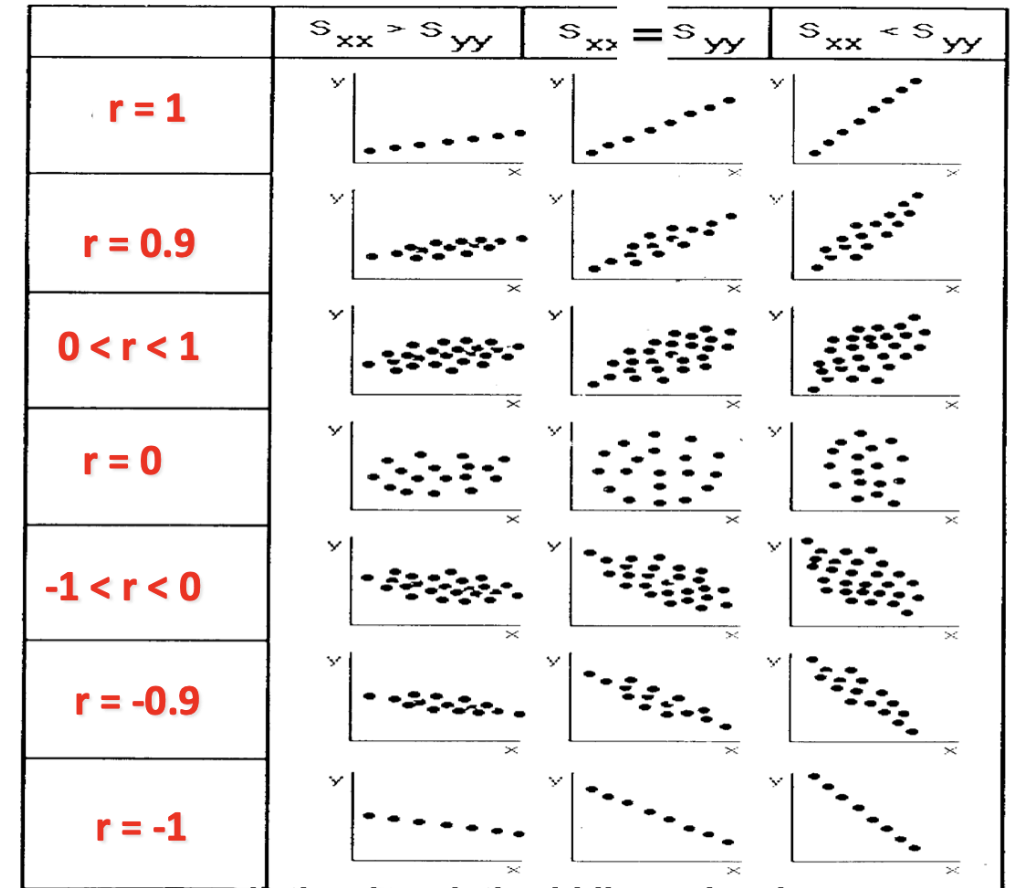
$$r_{X,Y} = \frac{\text{cov}_{X,Y}}{\sigma_x \cdot \sigma_y} \quad R = \begin{pmatrix} 1 & r_{12} \\ r_{21} & 1 \end{pmatrix}$$

The correlation coefficient between two variables X and Y is invariant under any linear transformation applied to the variables.

Linear correlation coefficient

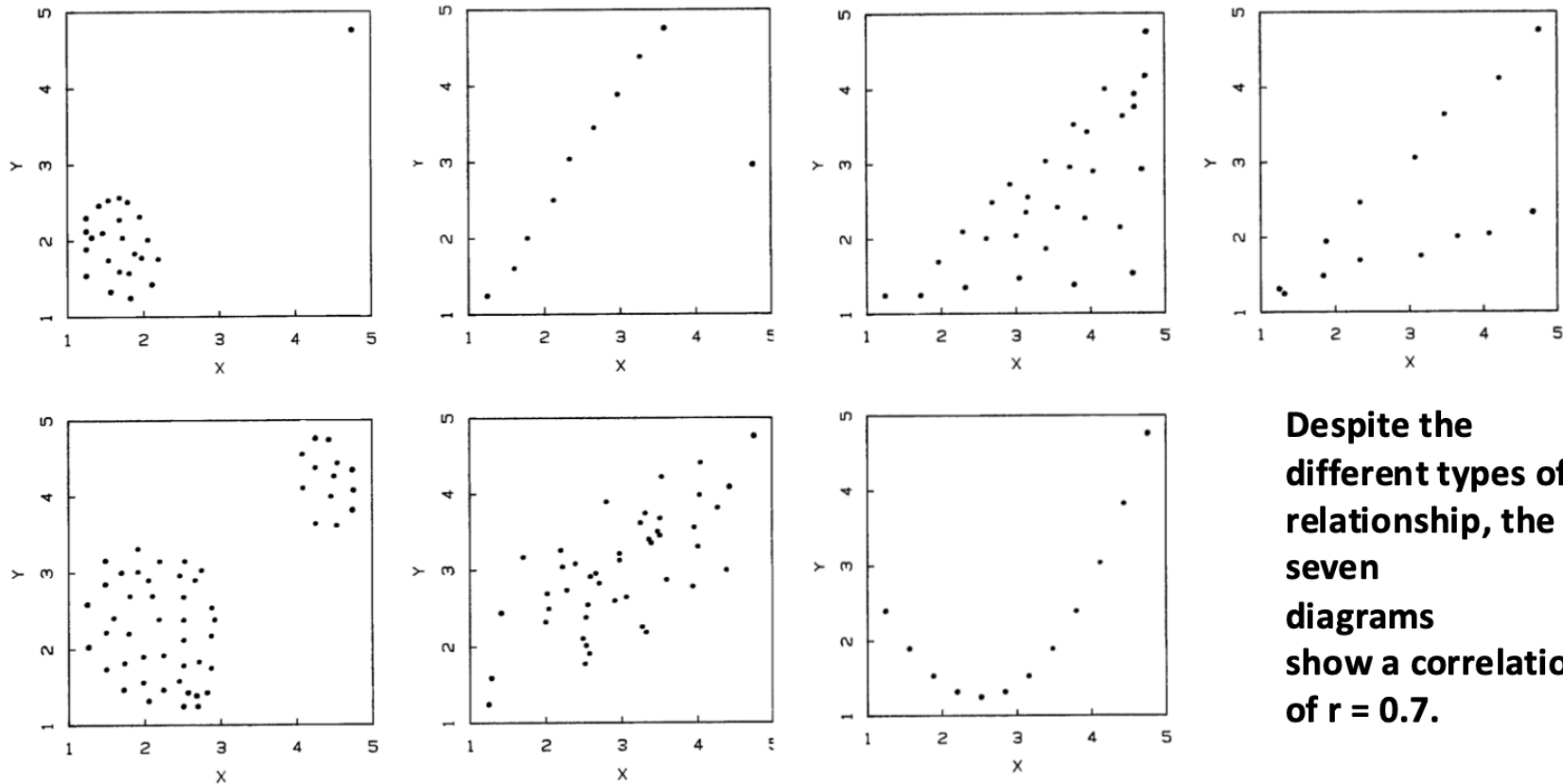
Properties:

- r_{xy} is always between -1 and +1, and is dimensionless.
- $|r_{xy}| = 1$ if, and only if, in the scatter plot the points are exactly located on a straight line (increasing if $r = +1$ or decreasing if $r = -1$)
- If two variables are independent, the correlation coefficient between them in the population is equal to 0.
- Therefore, the closer (in absolute value) r is to 1, the closer the linear relationship between X and Y.
- $100 \cdot r_{xy}^2$ can be interpreted as the percentage of the variability of Y that is “associated” with the variability of X



Linear correlation coefficient

The value of r_{xy} does not replace the information in the diagram.



Despite the different types of relationship, the seven diagrams show a correlation of $r = 0.7$.

Interpretation of relationships

Causal unidirectional dependency

Cause: Image resolution → Effect: Processing time

Partial dependence on one or more variables

Cause: Genetic characteristics → Effect: Height and Weight

Interdependence between the two variables

Supply and demand for a medication

Multiple Box-Whiskers

